VLSM (Variable Length Subnet Mask) Address Plan for 50 Rural Health Clinics

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Executive overview

This report sets out an IPv4 (Internet Protocol Version 4) address plan for 50 HealthReach clinics across Rivers, Benue and Kaduna states. All clinic addresses come from a single pool, 10.80.0.0/21, which is divided by region: Rivers takes 10.80.0.0/22, Benue takes 10.80.4.0/23 and Kaduna takes 10.80.6.0/23. Inside those regional blocks, the 10 large clinics get /26 networks, the 20 medium clinics get /27 networks and the 20 small clinics get /28 networks. Section 4 lists the network address, usable range, broadcast address and default gateway for every clinic.

In total, 1,600 addresses go to clinic subnets and the remaining 448 stay in reserve for growth inside the regional blocks. Because each region is a single aligned block, the cloud needs only three summary routes instead of 50 individual clinic prefixes. A clinic keeps the same addresses whether it is on cellular or satellite. Private addressing and route aggregation do the addressing work here, while encryption, access controls and local data storage handle privacy and offline operation separately.

The clinic counts per state, the device numbers and the hub addresses used throughout are my own planning assumptions, not figures given in the scenario. A site survey should confirm them before anything is deployed.

1 Scenario interpretation and planning assumptions

HealthReach Foundation provides vaccination and maternal-health services in three Nigerian states. Each clinic uses tablets and cold-chain sensors, which synchronise with a regional hub and a cloud dashboard over 3G/4G or satellite links. The task is to produce an efficient address plan. Because rural links drop out often, the plan also has to show how subnet allocation, routing and address management keep working through interruptions.

Assumption	Planning value and rationale
Clinic distribution	Rivers 20, Benue 15 and Kaduna 15, giving 50 in total. The split is illustrative because the scenario gives no state-level counts.
Clinic identifiers	RV01–RV20, BN01–BN15 and KD01–KD15 are planning labels, not real clinic names or locations.
Large clinics	10 clinics, each needing 50 addressed interfaces including the gateway and room for growth.
Medium clinics	20 clinics, each needing 24 addressed interfaces including the gateway and room for growth.
Small clinics	20 clinics, each needing 10 addressed interfaces including the gateway and room for growth.
Address scope	IPv4 clinic LANs only, using stable private addresses carried over routed VPNs. Carrier WAN addresses are not covered by this plan.
Hub and cloud scope	A separate 10.81.0.0/24 infrastructure pool, kept out of the clinic allocation and the regional summaries.
LAN boundary	One IPv4 subnet per clinic, as the schedule requires. No Layer 2 segment is shared between clinics.
Operational model	Local DHCP and local synchronisation storage stay available when the WAN is down.

1.1 Illustrative device budgets

Addressed interface category	Large	Medium	Small
Clinical tablets	24	10	4
IoT sensors or IP gateways	12	6	2
Local sync server	1	1	1
Other equipment management interfaces	5	3	1
Default gateway interface	1	1	1
Planned growth allowance	7	3	1
Total design demand	50	24	10

The budgets count IP interfaces, not people. A non-IP sensor sitting behind an IoT gateway does not need its own IPv4 address, but the estimate assumes each listed IoT device is addressed individually unless a gateway consolidates it. Guest Wi-Fi and extra security VLANs are left out. If either is needed, each must get its own subnet budget before deployment.

2 Addressing principles and VLSM calculations

VLSM (variable-length subnet masking) lets one address plan use subnets of different sizes. For each clinic I chose the smallest ordinary LAN subnet that fits its host requirement, and placed the larger subnets first so that every block stays properly aligned. CIDR aggregation then lets contiguous, aligned allocations be advertised as a single shorter prefix, which cuts the number of routes carried upstream (Fuller & Li, 2006).

The chosen pool sits inside the private 10.0.0.0/8 range defined in RFC 1918. These addresses are for the organisation's own network and must never be advertised on the public Internet. Other organisations can use the same ranges, so the cloud network, partner networks and VPN pools need to be checked for overlap before this plan is adopted (Rekhter et al., 1996).

2.1 Host sizing rule

On an ordinary IPv4 LAN, the number of usable hosts is 2^(32 − prefix length) − 2, because the network address and the directed broadcast address cannot be assigned to devices. Given a demand H, the aim is to find the smallest number of host bits h where 2^h − 2 ≥ H, and the prefix is then 32 − h. The demand figures already include the default gateway, so it is not added a second time.

Clinic class	Demand	Calculation	Prefix and mask	Usable	Unused usable
Large	50	$2^6 - 2 = 62$	/26 255.255.255.192	62	12
Medium	24	$2^5 - 2 = 30$	/27 255.255.255.224	30	6
Small	10	$2^4 - 2 = 14$	/28 255.255.255.240	14	4

A /27 cannot serve 50 interfaces because it offers only 30 usable addresses, and a /28 cannot serve 24 because it offers only 14. Likewise, a /29 falls short of 10 interfaces with just six usable addresses. The chosen prefixes are therefore the smallest conventional LAN sizes that meet the assumed demand. The /31 exception applies to point-to-point links and is not suitable for multi-access clinic LANs (Retana et al., 2000).

2.2 Allocation sequence and boundary checks

A /26 advances in steps of 64 addresses, a /27 in steps of 32 and a /28 in steps of 16. RV01, for example, is 10.80.0.0/26, with network address .0, usable range .1 to .62 and broadcast address .63. RV02 then starts at .64, so the two never overlap. After four /26 allocations, RV05 begins at 10.80.1.0/27, and after eight /27 allocations, RV13 begins at 10.80.2.0/28.

3 Regional hierarchy and route summaries

Region	Clinic mix (L/M/S)	Clinic addresses	Regional summary	Full covered range
Rivers	4 / 8 / 8	640	10.80.0.0/22	10.80.0.0–10.80.3.255
Benue	3 / 6 / 6	480	10.80.4.0/23	10.80.4.0–10.80.5.255
Kaduna	3 / 6 / 6	480	10.80.6.0/23	10.80.6.0–10.80.7.255
Total	10 / 20 / 20	1,600	10.80.0.0/21	10.80.0.0–10.80.7.255

Rivers needs 4 × 64 + 8 × 32 + 8 × 16 = 640 addresses. The smallest power-of-two block that covers this is 1,024 addresses, which is a /22. Benue and Kaduna each need 3 × 64 + 6 × 32 + 6 × 16 = 480 addresses, and a 512-address /23 is enough for each. Together the three regional blocks occupy 2,048 addresses, which exactly fills the /21 parent pool.

3.1 Why the summaries are valid

The Rivers summary uses the mask 255.255.252.0 and covers third-octet values 0 to 3. Benue and Kaduna use 255.255.254.0, covering values 4 to 5 and 6 to 7 respectively. The three blocks are aligned, do not overlap, and all sit inside 10.80.0.0/21, whose mask is 255.255.248.0. In binary, third octets 0 to 3 share their first six bits, 4 to 5 share seven, and 6 to 7 share seven. Third octets 0 to 7 share five, which gives the overall /21.

These summaries deliberately cover reserved space as well as installed clinics. They record who owns an address range and do not claim that every covered address is in use. No other region may allocate a subnet inside them. As a result, a summary stays stable when a single clinic link goes down, and more-specific routes handle reachability inside the region.

3.2 Spare regional blocks

Region	Unallocated prefixes	Addresses	Possible additional clinics
Rivers	10.80.2.128/25 10.80.3.0/24	384	6 large, or 12 medium, or 24 small
Benue	10.80.5.224/27	32	1 medium, or 2 small
Kaduna	10.80.7.224/27	32	1 medium, or 2 small
Total	Four reserved CIDR blocks	448	Capacity depends on regional demand and clinic mix

The expansion figures are alternatives, not amounts that can all be added at once. Benue and Kaduna cannot add a new /26 without reallocating space or being given another regional prefix. This is the trade-off between a tight initial allocation and growth flexibility. If site surveys point to faster expansion in either state, the better option is to reserve extra blocks outside this /21 and accept one more summary route, rather than stretching a summary into another region’s space.

4 Complete clinic subnet schedule

Every address is written out in full. The default gateway is the first usable address, and the usable range includes it. Prefix masks and host capacities are defined in Section 2. The last usable address is reserved for the local sync server, so once the gateway is assigned, /26, /27 and /28 LANs have 61, 29 and 13 usable addresses left respectively.

4.1 Rivers clinic allocations

Clinic	Network	Usable start and end	Broadcast	Gateway	Usable
RV01	10.80.0.0/26	10.80.0.1 10.80.0.62	10.80.0.63	10.80.0.1	62
RV02	10.80.0.64/26	10.80.0.65 10.80.0.126	10.80.0.127	10.80.0.65	62
RV03	10.80.0.128/26	10.80.0.129 10.80.0.190	10.80.0.191	10.80.0.129	62
RV04	10.80.0.192/26	10.80.0.193 10.80.0.254	10.80.0.255	10.80.0.193	62
RV05	10.80.1.0/27	10.80.1.1 10.80.1.30	10.80.1.31	10.80.1.1	30
RV06	10.80.1.32/27	10.80.1.33 10.80.1.62	10.80.1.63	10.80.1.33	30
RV07	10.80.1.64/27	10.80.1.65 10.80.1.94	10.80.1.95	10.80.1.65	30
RV08	10.80.1.96/27	10.80.1.97 10.80.1.126	10.80.1.127	10.80.1.97	30
RV09	10.80.1.128/27	10.80.1.129 10.80.1.158	10.80.1.159	10.80.1.129	30
RV10	10.80.1.160/27	10.80.1.161 10.80.1.190	10.80.1.191	10.80.1.161	30
RV11	10.80.1.192/27	10.80.1.193 10.80.1.222	10.80.1.223	10.80.1.193	30
RV12	10.80.1.224/27	10.80.1.225 10.80.1.254	10.80.1.255	10.80.1.225	30
RV13	10.80.2.0/28	10.80.2.1 10.80.2.14	10.80.2.15	10.80.2.1	14
RV14	10.80.2.16/28	10.80.2.17 10.80.2.30	10.80.2.31	10.80.2.17	14
RV15	10.80.2.32/28	10.80.2.33 10.80.2.46	10.80.2.47	10.80.2.33	14
RV16	10.80.2.48/28	10.80.2.49 10.80.2.62	10.80.2.63	10.80.2.49	14
RV17	10.80.2.64/28	10.80.2.65 10.80.2.78	10.80.2.79	10.80.2.65	14
RV18	10.80.2.80/28	10.80.2.81 10.80.2.94	10.80.2.95	10.80.2.81	14
RV19	10.80.2.96/28	10.80.2.97 10.80.2.110	10.80.2.111	10.80.2.97	14
RV20	10.80.2.112/28	10.80.2.113 10.80.2.126	10.80.2.127	10.80.2.113	14

4.2 Benue clinic allocations

Clinic	Network	Usable start and end	Broadcast	Gateway	Usable
BN01	10.80.4.0/26	10.80.4.1 10.80.4.62	10.80.4.63	10.80.4.1	62
BN02	10.80.4.64/26	10.80.4.65 10.80.4.126	10.80.4.127	10.80.4.65	62
BN03	10.80.4.128/26	10.80.4.129 10.80.4.190	10.80.4.191	10.80.4.129	62
BN04	10.80.4.192/27	10.80.4.193 10.80.4.222	10.80.4.223	10.80.4.193	30
BN05	10.80.4.224/27	10.80.4.225 10.80.4.254	10.80.4.255	10.80.4.225	30
BN06	10.80.5.0/27	10.80.5.1 10.80.5.30	10.80.5.31	10.80.5.1	30
BN07	10.80.5.32/27	10.80.5.33 10.80.5.62	10.80.5.63	10.80.5.33	30
BN08	10.80.5.64/27	10.80.5.65 10.80.5.94	10.80.5.95	10.80.5.65	30
BN09	10.80.5.96/27	10.80.5.97 10.80.5.126	10.80.5.127	10.80.5.97	30
BN10	10.80.5.128/28	10.80.5.129 10.80.5.142	10.80.5.143	10.80.5.129	14
BN11	10.80.5.144/28	10.80.5.145 10.80.5.158	10.80.5.159	10.80.5.145	14
BN12	10.80.5.160/28	10.80.5.161 10.80.5.174	10.80.5.175	10.80.5.161	14
BN13	10.80.5.176/28	10.80.5.177 10.80.5.190	10.80.5.191	10.80.5.177	14
BN14	10.80.5.192/28	10.80.5.193 10.80.5.206	10.80.5.207	10.80.5.193	14
BN15	10.80.5.208/28	10.80.5.209 10.80.5.222	10.80.5.223	10.80.5.209	14

4.3 Kaduna clinic allocations

Clinic	Network	Usable start and end	Broadcast	Gateway	Usable
KD01	10.80.6.0/26	10.80.6.1 10.80.6.62	10.80.6.63	10.80.6.1	62
KD02	10.80.6.64/26	10.80.6.65 10.80.6.126	10.80.6.127	10.80.6.65	62
KD03	10.80.6.128/26	10.80.6.129 10.80.6.190	10.80.6.191	10.80.6.129	62
KD04	10.80.6.192/27	10.80.6.193 10.80.6.222	10.80.6.223	10.80.6.193	30
KD05	10.80.6.224/27	10.80.6.225 10.80.6.254	10.80.6.255	10.80.6.225	30
KD06	10.80.7.0/27	10.80.7.1 10.80.7.30	10.80.7.31	10.80.7.1	30
KD07	10.80.7.32/27	10.80.7.33 10.80.7.62	10.80.7.63	10.80.7.33	30
KD08	10.80.7.64/27	10.80.7.65 10.80.7.94	10.80.7.95	10.80.7.65	30
KD09	10.80.7.96/27	10.80.7.97 10.80.7.126	10.80.7.127	10.80.7.97	30
KD10	10.80.7.128/28	10.80.7.129 10.80.7.142	10.80.7.143	10.80.7.129	14
KD11	10.80.7.144/28	10.80.7.145 10.80.7.158	10.80.7.159	10.80.7.145	14
KD12	10.80.7.160/28	10.80.7.161 10.80.7.174	10.80.7.175	10.80.7.161	14
KD13	10.80.7.176/28	10.80.7.177 10.80.7.190	10.80.7.191	10.80.7.177	14
KD14	10.80.7.192/28	10.80.7.193 10.80.7.206	10.80.7.207	10.80.7.193	14
KD15	10.80.7.208/28	10.80.7.209 10.80.7.222	10.80.7.223	10.80.7.209	14

As a worked example, BN10 is a small clinic on 10.80.5.128/28. Its gateway is 10.80.5.129, its sync server can use 10.80.5.142 and its broadcast address is 10.80.5.143. Its route belongs only to the Benue summary, 10.80.4.0/23.

5 Supporting hub and cloud address plan

The infrastructure pool is kept apart from the clinic pool so that the regional summaries stay easy to read. The allocations below are illustrative, and the actual cloud network and provider constraints must still be checked. Sync services and VPN router addresses are listed separately to make clear that an application server does not forward regional traffic.

Purpose	Subnet	Gateway or VPN router	Service example	Broadcast
Rivers hub	10.81.0.0/27	10.81.0.1	Sync 10.81.0.2	10.81.0.31
Benue hub	10.81.0.32/27	10.81.0.33	Sync 10.81.0.34	10.81.0.63
Kaduna hub	10.81.0.64/27	10.81.0.65	Sync 10.81.0.66	10.81.0.95
Cloud private services	10.81.0.128/26	10.81.0.129	Dashboard 10.81.0.130	10.81.0.191
Unallocated infrastructure	10.81.0.96/27 and 10.81.0.192/26	Not assigned	Reserved	As defined by each prefix

Some managed cloud subnets reserve more addresses than the usual network and broadcast pair, so the cloud service example needs adjusting to the chosen provider's rules. Public VPN endpoints, NAT addresses and SIM or satellite interface addresses are assigned by the provider, so none are invented here.

5.1 Optional numbered point-to-point tunnels

If the chosen equipment needs numbered tunnel interfaces, reserve 10.82.0.0/24 for them. One tunnel per clinic uses 50 /31 pairs (100 addresses), and two tunnels per clinic use 200. For primary tunnel index i from 0 to 49, allocate 10.82.0.(2i)/31, with the hub endpoint at 2i and the clinic endpoint at 2i + 1. Secondary tunnels use indices 50 to 99, which gives .100/31 through .198/31. This reserve is separate from the clinic and hub pools.

RFC 3021 allows both addresses in a /31 to be used on qualifying point-to-point links (Retana et al., 2000), but device support should be confirmed. If /30 prefixes are needed instead, two tunnels per clinic would take 100 × 4 = 400 addresses and call for at least a /23. Unnumbered or implementation-specific VPNs may not need dedicated tunnel addressing at all.

6 Regional routing and rural transport

6.1 Routing hierarchy

Each clinic router holds its directly connected LAN and reaches approved hub and cloud services through the active VPN. Each regional hub keeps individual routes to its own clinics and picks whichever clinic tunnel is currently working. The cloud VPN router holds just three clinic routes, one summary per regional hub. Return routes matter as much as outbound ones, since an outbound route alone does not make replies reachable.

Cloud route	Logical next hop	Scope
10.80.0.0/22	Rivers regional VPN peer	RV01–RV20 and Rivers reserve
10.80.4.0/23	Benue regional VPN peer	BN01–BN15 and Benue reserve
10.80.6.0/23	Kaduna regional VPN peer	KD01–KD15 and Kaduna reserve

The next hops are logical VPN peers rather than assumed public IP addresses. In a supported routed overlay, static routes with health tracking are enough for a 50-clinic design. A classless dynamic routing protocol could be added if the operations team has the skills to run it. Summaries must be configured explicitly at the regional boundary, and classful automatic summarisation should not be relied on.

6.2 Aggregate safety and failure handling

Each regional router should carry a discard route for its summary, with the preference set so that active, more-specific clinic routes win. Longest-prefix matching selects a /26, /27 or /28 before the /22 or /23 discard route. Traffic sent to reserved space or to a withdrawn clinic route is dropped locally, instead of following a default route back to the cloud and looping. CIDR aggregation needs this careful handling of unreachable covered destinations (Fuller & Li, 2006).

A region's summary should be advertised only while that region is an authorised and reachable attachment point. If a whole hub or its cloud tunnel fails, the matching cloud route should be withdrawn or deactivated using tunnel state and health checks. Surviving a hub failure needs a redundant hub or a deliberately engineered secondary path, because satellite backup at the clinics does not remove that dependency.

6.3 Cellular primary and satellite backup

The clinic LAN prefix stays the same on both transports. The clinic should start the authenticated VPN connection itself, so the design works with dynamic carrier addressing and carrier NAT where the provider and VPN software allow it. Reachability should be tracked to a hub or service beyond the local modem, since a registered SIM does not prove end-to-end connectivity. Traffic should normally prefer cellular and move to satellite only after a sustained failure, with a recovery delay to stop the link flapping.

Source NAT should be avoided inside the routed VPN where clinic identity needs to stay visible for logs and policy, although controlled NAT can be used for Internet-bound traffic at the chosen egress point. Tunnel MTU and TCP MSS should be set from measured path behaviour, because encryption overhead and different transports can cause fragmentation. The IP plan cannot add bandwidth, reduce satellite delay or guarantee operator coverage.

7 Privacy and offline resilience implications

7.1 Privacy controls around the address plan

Unique subnets make policy enforcement and incident tracing easier, but neither RFC 1918 addresses nor VLSM encrypt patient records. Clinic-to-hub traffic should use authenticated, encrypted VPNs, and clinical synchronisation should add application-level encryption. IPsec provides a network-layer security architecture, although the confidentiality actually achieved depends on the protection and algorithms configured (Kent & Seo, 2005).

Firewalls should deny by default. Clinic sync clients should reach only the designated hub or cloud service and the local services they need, clinic-to-clinic traffic should be blocked unless there is a reason for it, and management access should come only from authorised administrator networks. Each sensor or gateway should have its own credentials and restricted destinations, and tablets and locally stored patient data should be protected with access controls and encryption. Source IP addresses help with logging but are not reliable proof of who a person is.

The one-subnet-per-clinic schedule is the minimum the assignment requires, and on its own it does not separate sensors from tablets on the same LAN. Where supported, wireless client isolation, device firewalls or enforced switch policies can help. For stronger Layer 3 separation, add distinct clinical, IoT and management VLAN subnets after revising the device budget and allocation. The same subnet must not be reused across different VLANs, and unused host addresses should not be treated as a separately routable subnet.

7.2 Address administration during outages

DHCP should run on each clinic router instead of relying on a cloud DHCP server. The first usable address goes to the gateway and the last to the local sync server. Infrastructure devices and sensors should use recorded DHCP reservations, and the dynamic pool must exclude every static or reserved address. With local leases and a stable LAN, tablets can keep working locally when the WAN fails.

Local DNS records or a resilient service-discovery method should point to the sync server, and applications should not have cloud IP addresses hard-coded. Tablets can simply take their addresses from DHCP. An IP address management register should record, for each clinic, the ID, prefix, gateway, reservations, device identity, regional owner and approval history. Leases and utilisation should be reviewed periodically before any pool is expanded.

7.3 Application resilience

While offline, vaccination and maternal-health records should be stored securely on the tablet or the clinic server. Each record needs a stable identifier, and pending changes should be queued. When the connection returns, synchronisation should use acknowledgements, retries and duplicate detection, with clear conflict rules and an audit trail. Cold-chain gateways should buffer timestamped sensor readings and raise essential temperature alarms locally, since remote dashboards are unreachable during a WAN outage.

When satellite is the active link, alarms and small clinical updates should go ahead of large downloads. The queue should report the number of pending records, the age of the oldest one and the time of the last successful sync. Enough local storage, backup power and reliable timekeeping are also implementation requirements. These are design recommendations drawn from the scenario, not capabilities that subnetting provides.

8 Efficiency assessment

Metric	Result	Interpretation
Allocated clinic subnet addresses	1,600	10 × 64 + 20 × 32 + 20 × 16
Usable clinic host addresses	1,500	1,600 less 100 network and broadcast addresses
Design demand including gateways	1,180	10 × 50 + 20 × 24 + 20 × 10
Unused usable capacity	320	Headroom inside active clinic subnets
Host capacity utilisation	78.67%	1,180 ÷ 1,500 × 100
Clinic allocation share of parent	78.125%	1,600 ÷ 2,048 × 100
Unallocated regional reserve	448	2,048 − 1,600
Upstream clinic route reduction	94%	(50 − 3) ÷ 50 × 100

These metrics measure different things. The 320 unused usable addresses sit inside existing clinic subnets and cannot be given to new clinics without resubnetting. The 448 reserved addresses lie outside those subnets and can be divided into new networks. Since the assumed demand already includes a growth allowance, 78.67% is planned capacity utilisation, not utilisation measured on live devices.

8.1 Comparison with uniform addressing

Giving every clinic a /26 would use 3,200 addresses, against 1,600 for the VLSM clinic subnets, a saving of 50% at the subnet level. A /24 per clinic would use 12,800 addresses, so VLSM saves 87.5% compared with that. Uniform /27 or /28 allocations would be too small for the larger clinics. These comparisons leave out regional aggregate padding in every design, and they describe address space only, not traffic or bandwidth savings.

The three summary routes reduce the clinic entries on the cloud router, but the regional hubs still need individual clinic routes. If a higher private core reaches every region through one cloud VPN attachment, that attachment can advertise 10.80.0.0/21 upstream. A router that must choose between the three regional peers needs to keep the three regional summaries, because a single /21 would hide the next-hop differences it depends on.

9 Validation and implementation procedure

Before deployment, the inventory should be verified and the plan checked for overlaps with existing networks. If actual demand in any clinic category exceeds the assumption, the clinic prefix should be resized while keeping regional alignment, and every affected schedule entry revised. The largest subnets should be allocated first, and unallocated prefixes recorded as reserved in the address register.

Validation item	Result for this plan
Clinic count	20 + 15 + 15 = 50; all IDs are distinct.
Subnet overlap	All 50 clinic networks are pairwise disjoint.
Boundary alignment	Every /26, /27 and /28 begins on the correct block boundary.
Gateway validity	Each gateway is the first usable address and is neither network nor broadcast.
Demand capacity	Each subnet meets its class demand of 50, 24 or 10 interfaces.
Regional containment	Every clinic subnet belongs to exactly its region summary.
Parent containment	All regional summaries fit 10.80.0.0/21 without overlap.
Infrastructure separation	10.81.0.0/24 and 10.82.0.0/24 do not intersect clinic space.
Reserve accounting	640 + 480 + 480 + 448 = 2,048 addresses.

9.1 Deployment sequence

1. Approve the surveyed clinic counts and device budgets, and check existing LAN, cloud and partner VPN address ranges for overlap.
2. Record the parent pool, regional summaries, clinic prefixes and reserves in the central address register.
3. Configure each clinic gateway, local DHCP, reservations and local sync service, and verify the masks and exclusions.
4. Configure the encrypted primary and backup VPNs, hub routes to each clinic, cloud return routes, filtering and aggregate discard routes.
5. Test local operation, cross-subnet access policy and full two-way clinical synchronisation.
6. Disconnect cellular service and confirm satellite failover without renumbering the LAN. Then disconnect both links and confirm that local data capture and sensor alarms still work.
7. Restore connectivity and verify that queued records synchronise exactly once, with no unexplained losses or duplicates. Confirm that routes are withdrawn when a whole hub fails.
8. Hand over the address register, recovery procedures and monitoring thresholds, and review address use and regional reserves on a regular schedule.

9.2 Acceptance criteria

A clinic passes when its local devices receive only valid addresses, its gateway and sync server stay reachable offline, the intended services work over either transport, unauthorised inter-clinic access is denied and return routing is correct. In the normal topology the cloud must show exactly the three intended regional clinic routes, and traffic to an unused reserved prefix must not loop. Offline and failover tests need to show that the application keeps working, not just that a ping succeeds.

10 Conclusion

The plan delivers the required 50-clinic VLSM schedule using a 10.80.0.0/21 parent pool, three regional aggregates and /26, /27 and /28 clinic networks sized to the assumed demand. It pairs minimum subnet sizing with explicit regional reserves and keeps addresses stable across cellular and satellite links. Its main limitations are the assumed inventory, the limited growth space in Benue and Kaduna, and the single-subnet baseline at each clinic. These assumptions should be confirmed, and the separate privacy and offline controls implemented, before the network is put into operation.assumed demands with explicit regional reserves and stable private addresses across cellular and satellite links. Its principal limitations are the assumed inventory, limited Benue and Kaduna growth space, and the single-subnet baseline at each clinic. Confirming those assumptions and implementing the separate privacy and offline controls are necessary before operational use.

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